# A/B Testing Interview Playbook

Framework · Metric types and tests · Sample-size math · 6 worked projects · 30-question bank

| Part | What's inside |
|---|---|
| 1 | The 8-step framework (what to say, what to avoid) |
| 2 | Metric types: binary, continuous, ratio, skewed, retention, and which test fits |
| 3 | Sample-size math: intuition, formulas, one worked number each, split cost, CUPED, clusters, duration |
| 4 | Six practice projects: retail coupon · Groww SIP nudge · Hike Cinema Continue Watching · notification timing · seller fee change (marketplace) · Netflix home page and price increase |
| 5 | Question bank with crisp answers |
| 6 | 60-second answer script and final checklist |

**How to use it:** Pick a project, cover the answer, and say the design out loud in the order of the framework. Then check the numbers. All sample sizes assume two-sided $\alpha = 0.05$ and power $= 0.80$ unless stated, and were computed with the formulas in Part 3. Every assumption (baseline, $\sigma$, traffic) is invented for practice: in a real interview, state your assumptions and ask the interviewer for real ones.

# Part 1 · The framework

Say these in order. Each step has a one-line "say it" and a common trap. The ordering itself signals seniority: decision first, metrics second, statistics third.

| # | Step | Say it like this | Common trap |
|---|---|---|---|
| 1 | Decision | "What will we do with the result: launch, iterate, or stop? Is a test even worth running?" | Jumping to randomization or metrics |
| 2 | Hypothesis | "If we show X to Y users, then Z changes, because [mechanism]." | Vague: "it will improve engagement" |
| 3 | Metrics | "One primary metric (defined: numerator, denominator, window), 2–3 diagnostic metrics, 2–4 guardrails." | Several primaries, chosen after seeing data |
| 4 | Eligibility, unit, entry | "Eligible users are… Randomize at user/account level so one person sees one variant. Users enter the experiment at first exposure, not at app open." | Dilution (counting users who never saw it); session-level randomization |
| 5 | Sizing | "α 0.05 two-sided, power 0.8, baseline p, MDE δ (a business call) → n per arm → split → days." | Splitting 50/50 by default without checking cost; MDE picked for statistical convenience |
| 6 | Duration | "n ÷ daily eligible entrants, rounded up to ≥ 2 full weeks, plus the outcome window. No peeking." | Stopping when p first dips below 0.05 |
| 7 | Validity checks | "SRM check first. Then novelty, seasonality, other tests, logging." | Reading results before checking SRM |
| 8 | Analysis + decision rules | "Effect size with CI, guardrail deltas, segments. Pre-commit: win → launch, mixed → iterate, flat → stop." | Reporting p-value only; changing the success metric after the result |

## Three habits that separate senior answers

- **Lead with the decision and the trade-off**, not the t-test.
- **Name the failure mode** before the interviewer does: selection bias, novelty, metric gaming, interference, pull-forward.
- **Use a concurrent control with intent-to-treat (ITT)**: compare everyone assigned to treatment against everyone assigned to control, never "users who used the feature" vs. the rest.

# Part 2 · Metric types, tests, and how to size

The metric's **type** decides the test, the sample-size formula, and the failure modes. Say the type out loud before you pick the test.

| Type | Examples | Test | Sizing and watch-outs |
|---|---|---|---|
| **Binary** (yes/no per user) | Conversion, coupon redeemed, D7 retained, SIP started | Two-proportion z-test (≡ chi-square on 2×2). Report absolute lift and CI; logistic regression if adjusting for covariates | $n = Z^2 [p_1q_1 + p_2q_2] / \delta^2$. Denominator = *exposed eligible users*. Low baselines (1–2%) need large n |
| **Continuous** (mean per user) | Revenue/user, watch minutes, basket value | Welch's t-test (no equal-variance assumption) | $n = 2Z^2\sigma^2/\delta^2$. Usually right-skewed: use CUPED, cap at the 99th percentile (report both), or bootstrap the CI |
| **Zero-inflated continuous** | Revenue where most users pay 0 | Decompose: P(paid) × E[amount \| paid], or test the overall mean with CUPED | Two effects can offset (more payers, smaller baskets). Look at both |
| **Ratio** (numerator ÷ denominator) | CTR = clicks / impressions, revenue / session | Delta method, or aggregate to the randomization unit (per-user CTR) | Randomize by user but divide by sessions → naive SE is too small → false positives |
| **Count** | Sessions/week, orders/user | t-test on the mean is fine at large n; Poisson/negative-binomial regression if needed | Cap outliers (bots, power users) |
| **Retention / time-to-event** | D30 retention, churn, time to 2nd purchase | Fixed horizon: binary test. Whole curve: Kaplan–Meier + log-rank, or Cox | Users must have the full window; censoring. Define "active" as an action with intent (message sent), not an app open |
| **Clustered / geo** | Seller groups, cities, stores | Cluster-robust SE, DiD, synthetic control | Design effect $DE = 1 + (m-1)\cdot ICC$ shrinks effective n massively |

## Metric roles

| Role | Purpose | Rule |
|---|---|---|
| Primary (OEC) | The one metric the ship decision rests on | Exactly one; pre-registered; closest to the intended user outcome |
| Diagnostic | Explains *why* the primary moved (CTR, funnel steps) | Not used to declare a win |
| Guardrail | Must not get worse (refunds, cancellations, opt-outs, latency, churn) | Pre-set thresholds. A breached guardrail blocks the launch even when the primary wins |
| Long-term / habit | Does the gain persist? (day-60 sustained use, retention) | Follow users on a holdout beyond the readout |

**Interview line:** "Binary outcome → proportions test. Continuous → Welch's t. Skewed revenue → CUPED or capping so a few whales don't drive the result. Ratio metrics → compute at the randomization unit or use the delta method."

# Part 3 · Sample-size math

## Intuition

A test is a signal-to-noise problem. The **signal** is the lift you want to detect (δ, the MDE). The **noise** is the metric's variance ($\sigma^2$). You need enough users that the noise averages out below the signal. So n grows with variance and shrinks with the **square** of the lift: halve the MDE and you need about 4× the users.

## Building blocks

| Parameter | Typical value | Meaning |
|---|---|---|
| α (two-sided) | 0.05 → Z = 1.96 (0.01 → 2.576) | False-positive rate: shipping a dud |
| Power (1 − β) | 0.80 → Z = 0.84 (0.90 → 1.28) | Chance of detecting a real effect of size δ |
| $(Z_{\alpha/2} + Z_\beta)^2$ | $(1.96 + 0.84)^2 = 7.85$ (power 0.9: 10.51) | Constant that goes into every formula |
| MDE (δ) | A business decision | Smallest lift that would change the roadmap or pay back the cost |
| Baseline, σ | From historical data | Variance of a binary metric = p(1 − p) |

## Formulas (per arm, equal split)

| Metric | Formula | Rule of thumb |
|---|---|---|
| Binary | $n = (Z_{\alpha/2} + Z_\beta)^2 \cdot [p_1(1-p_1) + p_2(1-p_2)] / (p_2 - p_1)^2$ | $n \approx 15.7 \cdot p(1-p) / \delta^2$ |
| Continuous | $n = 2 (Z_{\alpha/2} + Z_\beta)^2 \cdot \sigma^2 / \delta^2$ | $n \approx 15.7 \cdot \sigma^2 / \delta^2$ |

### Worked number 1: binary

```
Baseline purchase rate 8%, want to detect +1pp (to 9%).
p1(1-p1) = 0.08 x 0.92 = 0.0736;  p2(1-p2) = 0.09 x 0.91 = 0.0819;  sum = 0.1555
n = 7.85 x 0.1555 / (0.01)^2 = 7.85 x 0.1555 / 0.0001 = 12,206 per arm
```

### Worked number 2: continuous

```
Revenue/customer: sd = Rs.900, want to detect +Rs.20.
n = 2 x 7.85 x 900^2 / 20^2 = 15.7 x 810,000 / 400 = 31,788 per arm
```

## Three levers that change n

- **MDE:** same 8% baseline, detect +0.5pp instead of +1pp → 47,525 per arm vs 12,206 (≈ 3.9×).
- **Variance reduction (CUPED):** adjust the metric with its pre-experiment value. Variance falls by $(1 - \rho^2)$. With $\rho = 0.5$, the revenue test above drops from 31,788 to 23,841 per arm (−25%).
- **Power and α:** power 0.9 needs 10.51/7.85 ≈ 1.34× the users; α = 0.01 needs ≈ 1.5×.

## What an unequal split costs

Groww-style example: baseline 2%, MDE +0.5pp. For a treatment share $r = n_T/n_C$, control size is $n_C = Z^2 \cdot [p_C q_C + p_T q_T / r] / \delta^2$.

| Split (T : C) | n control | n treatment | Total | vs 50/50 |
|---|---|---|---|---|
| 50 : 50 | 13,807 | 13,807 | 27,614 | 1.00× |
| 20 : 80 | 36,765 | 9,192 | 45,957 | 1.66× |
| 10 : 90 | 75,028 | 8,337 | 83,365 | 3.02× |

Use skewed splits only to limit risk or cost (e.g. an expensive coupon, a risky change) and say so. 50/50 is the most efficient.

## Clusters (geo, seller groups, households)

When you randomize groups, users inside a group are correlated, so you have fewer "effective" users. **Design effect $DE = 1 + (m - 1) \cdot ICC$**, where m = users per cluster and ICC = intra-cluster correlation. Required users = n × DE. Example: ICC 0.05 and m = 200 → DE = 10.95, an 11× inflation.

## Duration

```
days = (n_control + n_treatment) / daily eligible NEW entrants
then: round UP to full weeks, minimum 2 weeks (weekday/weekend and pay-cycle effects)
```

```
then: add the outcome window (e.g. +7 days for a 7-day conversion; +60 days for habit)
note: the eligible pool depletes (users enter once), so use the decaying entrant curve, not a flat average.
```

## Holdout sizing for ongoing programs (coupons, campaigns)

Withholding the treatment costs revenue, so size the holdout at the *smallest* value that still hits n for the MDE, then keep a small permanent holdout (1–5%) to track long-term effects.

## Calculator

```
from scipy.stats import norm
import math
Z2 = (norm.ppf(.975) + norm.ppf(.80))**2

def n_binary(p1, p2, r=1.0):          # r = n_treat / n_control
    nc = Z2*(p1*(1-p1) + p2*(1-p2)/r)/(p2-p1)**2
    return math.ceil(nc), math.ceil(nc*r)

def n_cont(sd, delta, r=1.0, rho=0.0):  # rho = CUPED correlation
    nc = Z2*sd**2*(1-rho**2)*(1+1/r)/delta**2
    return math.ceil(nc), math.ceil(nc*r)

print(n_binary(.08, .09))   # (12206, 12206)
print(n_cont(900, 20))      # (31788, 31788)
```

# Project 1 · Retail coupon promotion (grocery chain)

## Context and assumptions

- 2M loyalty customers; plan: 15% off next purchase (cap ₹150), unique code sent by app/SMS.
- 14-day purchase rate: 8%. Revenue per customer over 14 days: mean ₹450, σ ₹900 (heavily right-skewed). Gross margin 20%.
- Coupon cost is incurred only on redemption. Business question: *is the coupon incremental, and is it profitable?*

## Hypothesis

If we send a 15%-off coupon to loyalty customers, 14-day gross margin per customer net of coupon cost rises, because the discount pulls forward and adds purchases; we must measure *incremental* margin versus a no-coupon holdout.

## Metrics

| Role | Metric | Type | Definition |
|---|---|---|---|
| Primary | Incremental margin per targeted customer, net of coupon cost | Continuous | (Revenue × 20% margin − coupon discount) per customer assigned, 14 days |
| Diagnostic | 14-day purchase rate | Binary | Customers with ≥1 purchase / customers assigned |
| Diagnostic | Basket value, items per basket | Continuous | Among purchasers, and also per assigned customer |
| Guardrail | Pull-forward: revenue days 15–42 | Continuous | If it drops below control, the lift was just timing |
| Guardrail | Full-price purchase share, margin % | Binary/Cont. | Detects discount dependency |

## Design

| Item | Choice |
|---|---|
| Unit | Customer (loyalty ID), one coupon per customer; unique codes to prevent sharing/leakage |
| Eligibility / entry | Active in last 90 days; enters at send date |
| Split | Large population → small control. 99:1 (1% holdout = 20k) vs. a 5% holdout = 100k: pick the smallest holdout that is still powered |
| Analysis | ITT: all assigned, redeemers or not. Welch t-test (CUPED on pre-period spend), plus z-test on purchase rate |
| Duration | Single send, 14-day window, then follow to day 42 for pull-forward |

## Sample-size math

```
Binary (purchase rate 8% → 9%):  n = 12,206 per arm
Continuous (revenue, sd 900, delta Rs.20):  n = 31,788 per arm   <- binding (needs the larger n)
With CUPED (rho = 0.5):  n = 23,841 per arm
Holdout at 99:1:  n_control needed = 16,055  -> a 1% holdout (20,000) is enough.
Break-even: incremental margin per customer must exceed average coupon cost per customer (assigned).
```

## Guardrails

- **Coupon cost vs. margin:** redemption rate is vanity; incremental margin is the decision metric.
- **Discount dependency:** track full-price purchases and later-period spend. Heavy couponing trains customers to wait.
- **Coupon leakage** (shared codes) contaminates control: use unique, non-transferable codes.

## Decision rules (pre-committed)

| Outcome | Action |
|---|---|
| Margin net of cost > 0 and no pull-forward | Launch; next step: uplift model to target persuadables |

| Outcome | Action |
|---|---|
| Purchase rate up but margin ≤ 0 | Don’t ship as-is: lower the discount, add a minimum basket, or target narrower |
| Lift disappears days 15–42 | It was pull-forward: reject or re-time |

## Likely follow-ups and strong answers

| Question | Answer |
|---|---|
| Why not compare redeemers vs. non-redeemers? | Selection bias: redeemers are your most engaged customers. Only assignment-based (ITT) comparison is causal. |
| Redemption is 12%: is the coupon working? | Unknown. Many redeemers would have bought anyway (subsidized intent). Incrementality = treatment minus holdout. |
| How do you pick the holdout size? | Smallest holdout that still powers the MDE; each extra control user is foregone coupon revenue. Keep 1–5% permanently for long-run effects. |
| How did you handle this at Flipkart? | Tie to your Meta test: an 8% holdout sized from power analysis, incrementality at BU level, and ROI = incremental GMV vs. spend. Same logic, different treatment. |
| What comes after? | Uplift modelling (T-/X-learner) on the holdout data to target persuadables, evaluated by a Qini curve. |

# Project 2 · Groww: ₹100 SIP nudge

## Context and assumptions

- Eligible: users who made one lump-sum investment but have no active SIP. About 20,000 new eligible users enter per day.
- Baseline 7-day SIP start = 2%; day-60 sustained SIP (≥ 3 installments, still active) = 1.2%.
- Sensitive domain (financial product, regulation): the real risk is low-quality SIPs that bounce or get cancelled.

## Hypothesis

If we show a “Start a SIP with ₹100” nudge on the home screen to single-investment users, SIP starts rise because a low ticket removes the commitment barrier. We also hypothesize that incremental users sustain the SIP (habit), which we test as a co-primary.

## Metrics

| Role | Metric | Type | Definition |
|---|---|---|---|
| Primary | SIP start rate, 7 days | Binary | Users with SIP mandate set up and first installment paid within 7 days of entry / eligible users exposed |
| Co-primary | Sustained SIP, day 60 | Binary | ≥ 3 installments paid and SIP active at day 60 / exposed eligible users (ITT) |
| Diagnostic | Nudge CTR; nudge → mandate completion; median SIP amount | Binary/Cont. | Where users drop in the funnel |
| Guardrail | Mandate failure/bounce rate; 30-day cancellation | Binary | Quality of the SIPs created |
| Guardrail | Lump-sum investment per user; uninstalls; complaints | Cont./Binary | Cannibalization and trust |

## Design

| Item | Choice |
|---|---|
| Unit | user_id (one user, one variant across devices) |
| Entry | First home-screen view as an eligible user, not app install (avoids dilution) |
| Split | 20 : 80 (treatment : control) for risk containment; costs ≈ 1.7× traffic vs 50/50. Say that explicitly |
| Analysis | Two-proportion z-test, absolute lift in pp with 95% CI. SRM chi-square against 20:80 first |
| Duration | ≥ 14 days (covers SIP dates 1st–10th), then 60-day follow-up per user with the holdout intact |

## Sample-size math

```
Start rate 2% → 2.5% (+0.5pp):
  50/50: 13,807 per arm = 27,614 total
  20/80: control 36,765, treatment 9,192 = 45,957 total (1.66x)
Day-60 sustained 1.2% → 1.5% (+0.3pp): 23,225 per arm at 50/50  <- binding
Traffic: ~20,000 entrants/day -> ~1.4 days for 50/50 start metric, so the 14-day floor binds, not n.
```

## Guardrails

- SIP bounce/failure rate and 30-day cancellation: a nudge that creates SIPs that die is not a win.
- Lump-sum cannibalization: are users swapping one-time buys for ₹100 SIPs?
- Regulatory/mis-selling complaints; the nudge must not mislead on risk.

## Decision rules (pre-committed)

| Outcome | Action |
|---|---|
| Start rate ≥ MDE, sustained SIPs up, guardrails flat | Launch; keep a 5% holdout for long-term persistence |
| Starts up, cancellations/bounces up | Don’t ship. Find the cause; try a better-fit segment or copy; rerun |

| Outcome | Action |
|---|---|
| Starts up but day-60 sustained flat | Gain was non-habitual: iterate on the nudge/follow-up journey |
| Flat or negative | Stop; check power first (don’t extend to chase significance) |

## Likely follow-ups and strong answers

| Question | Answer |
|---|---|
| Why not compare retention among starters to historical SIP retention? | Selection and mix: the nudge brings marginal, lower-intent users so starter-retention looks worse even if the nudge is net positive. Compare sustained SIPs per 1,000 exposed users, treatment vs. concurrent control. |
| Starts +0.8pp (p=0.01) but 30-day cancellations doubled. Ship? | No. Quantify net value: incremental sustained SIPs minus cancellations, AUM per SIP, and regulatory risk. Recommend iterate and rerun with the cancellation guardrail gating. |
| Why two-sided α? | Harm matters (nudge fatigue). One-sided is blind to a negative effect and is only defensible if pre-registered. |
| Market crashes mid-test? | Seasonality/regime shift hits both arms equally (concurrent control), but it can change the effect size: report by week and flag it. |

# Project 3 · Hike Cinema: “Continue Watching” row on home

## Context and assumptions

- Eligible: users with ≥ 1 unfinished episode. Micro-drama platform inside a messenger.
- Weekly watch minutes per eligible user: mean 40, σ 60 (right-skewed). Pre-period watch time correlates ρ = 0.6 with the outcome.
- Watch-ep-2-within-48h baseline 35%. Daily eligible entrants ≈ 15,000.

## Hypothesis

If we show a Continue Watching row at the top of home to users with an unfinished episode, 7-day watch time rises because resuming removes the search effort, without hurting return or notification opt-outs.

## Metrics

| Role | Metric | Type | Definition |
|---|---|---|---|
| Primary | Watch minutes per eligible user, 7 days | Continuous | Total minutes in 7 days after first exposure / eligible exposed users (cap at p99; report uncapped too) |
| Diagnostic | Watched ep 2 within 48h | Binary | Habit-loop proxy for serial content |
| Diagnostic | Row CTR; resume-to-complete rate | Ratio/Binary | User-level CTR (clicked at least once) |
| Guardrail | D7 return; notification opt-out; uninstall | Binary | Don’t trade long-term health for a watch-time spike |
| Guardrail | Messenger DAU/messages per user | Cont. | Cross-product health |

## Design

| Item | Choice |
|---|---|
| Unit / entry | user_id; entry at first home-screen render with an unfinished episode (so every counted user could see the row) |
| Split | 50 : 50 |
| Analysis | Welch t-test on the CUPED-adjusted capped metric; bootstrap CI as a robustness check; z-test on binary diagnostics |
| Duration | ≥ 2 weeks (novelty washes out), 7-day outcome window per user |

## Sample-size math

```
Continuous: sd 60, MDE 2 min (5%):  n = 14,128 per arm
With CUPED rho = 0.6 (variance x 0.64):  n = 9,042 per arm
Binary diagnostic (ep2 within 48h, 35% -> 37%):  n = 9,038 per arm
Days (50/50, CUPED): 18,084 / 15,000 per day = 1.2 days -> 14-day floor binds.
```

## Guardrails

- Watch time alone is gameable (autoplay drift): pair with completion and D7 return.
- Check the effect is stable across weeks; a decaying lift means novelty.
- Segment by new vs. returning, genre, language, and acquisition cohort.

## Decision rules (pre-committed)

| Outcome | Action |
|---|---|
| Watch time up, D7 return flat or up | Launch; monitor with a 5% holdout |
| Watch time up, return/opt-out worse | Passive watching: don’t ship; redesign (e.g. less aggressive autoplay) |
| Flat | Stop; consider a ranking change instead of placement |

## Likely follow-ups and strong answers

| Question | Answer |
| --- | --- |
| Why is watch time a risky primary? | It’s an output; optimizing it can mean passive or irrelevant viewing. Prefer intentional watch time, completions, and next-day return as co-metrics. |
| Revenue/watch-time is skewed: how to handle? | CUPED, cap at p99 (report both), bootstrap CI, and check the result isn’t driven by the top 0.1% of users. |
| Metric decays after week 1? | Novelty effect: read the steady-state weeks, extend or hold out, and report the stabilized effect. |
| Unit of randomization? | User, so a person sees one experience; entry at first eligible exposure to avoid dilution. |

# Project 4 · Notification timing (personalized send-time)

## Context and assumptions

- Messenger app. Treatment: send each user their push at their personal peak hour; control: current fixed schedule. Same notification volume.
- Week-4 retention (user sent ≥ 1 message in week 4) baseline 30%. Notification opt-out baseline 8%.
- Lesson from the case bank: “active = opened app” can be gamed by timing; use a message-sent definition.

## Hypothesis

If we send notifications at each user’s personal peak time, week-4 message-sender retention rises because relevance and timeliness increase engagement, without raising opt-outs.

## Metrics

| Role | Metric | Type | Definition |
|---|---|---|---|
| Primary | Week-4 message-sender retention | Binary | Users who sent ≥ 1 message in week 4 / users exposed in week 0 |
| Diagnostic | Sessions/user/day; push open rate | Count/Ratio | Mechanism checks; open rate computed per user |
| Guardrail | Notification opt-out rate | Binary | Over-messaging signal: **this one needs the largest n** |
| Guardrail | Uninstalls; messages sent per user | Binary/Cont. | Passive-opener detection |

## Design

| Item | Choice |
|---|---|
| Unit | user_id; graph clusters for messaging if peer spillover matters (friends receive treated users’ messages) |
| Entry | First notification-eligible day |
| Split | 50 : 50 |
| Analysis | Two-proportion z-test for retention and opt-out; SRM check; segment by tenure and platform |
| Duration | 2 weeks of enrollment + 4-week outcome window ⇒ ~6 weeks to read week-4 retention |

## Sample-size math

```
Retention 30% -> 31% (+1pp):  n = 33,272 per arm
Opt-out 8% -> 8.5% (+0.5pp, harm to detect):  n = 47,525 per arm  <- binding
Insight: guardrails often need more users than the primary because the harm you must detect is small.
```

## Guardrails

- Opt-out rate (the binding guardrail) and uninstall rate.
- Messages sent per user and sessions: confirms retention is real, not notification-induced opens.
- Network spillover: treated users change what their friends see.

## Decision rules (pre-committed)

| Outcome | Action |
|---|---|
| Retention up, opt-out and messages flat | Launch with holdout |
| Retention up (opens) but messages/sessions down, opt-out up | Metric gaming: don’t ship; fix the metric definition and rerun |
| Flat | Stop |

## Likely follow-ups and strong answers

| Question | Answer |
| --- | --- |
| D30 retention up, sessions down, opt-outs up. Ship? | No. “Retained” is app-open based and the treatment is gaming the metric. Judge on messages sent and opt-out; recommend fix and rerun. |
| Which sample size binds? | The opt-out guardrail, not the primary. Size on the harder one. |
| Network effects here? | Yes: friends of treated users are affected. Cluster/graph randomization (less power) or accept bias and state it. |
| Link to your own work | Festive churn model + random test/control on at-risk users. The holdout isolates incremental lift; the extension is uplift modelling. |

# Project 5 · Marketplace: seller fee discount (eBay-style)

## Context and assumptions

- Treatment: 2pp lower take rate for sellers in a category cluster. Sellers and buyers share listings, so a seller-level A/B contaminates control (interference, SUTVA violation).
- Metric: seller listing-activation rate (≥ 1 new listing in 30 days) baseline 30%. GMV per seller: mean ₹50k, σ ₹150k.
- Clusters: category × region groups, ~200 sellers each. ICC = 0.05. Only 40 comparable clusters may exist.

## Hypothesis

If we cut the take rate by 2pp for sellers in selected clusters, listing activation rises by ≥ 3pp because a lower fee improves seller economics, and platform net revenue (fees plus ad/promotion revenue) does not fall.

## Metrics

| Role | Metric | Type | Definition |
|---|---|---|---|
| Primary | Seller listing-activation rate | Binary | Sellers with ≥ 1 new listing in 30 days / sellers in cluster |
| Co-primary | Net revenue per seller | Continuous | Fees + ads − discount cost per seller, 60 days |
| Diagnostic | GMV, listings per seller, ASP, buyer conversion | Continuous | Decomposition: sellers × listings × conversion × ASP |
| Guardrail | Seller churn; buyer experience; cross-cluster spillover | Binary/Cont. | Supply health and contamination |

## Design

| Item | Choice |
|---|---|
| Unit | Cluster (category × region), not seller or buyer, to avoid interference |
| Design | Matched-pair randomization on pre-period GMV and growth. If few clusters: DiD or synthetic control |
| Analysis | Cluster-robust SE, DiD with pre-period trends check (parallel trends), placebo test |
| Duration | ≥ 4 weeks (sellers adjust slowly) + 60-day follow-up; avoid peak season |

## Sample-size math

```
Individual-level n for 30% -> 33%: 3,760 per arm
Design effect (ICC 0.05, m = 200): DE = 1 + 199 x 0.05 = 10.95
Sellers needed per arm = 3,760 x 10.95 = 41,173  ->  about 206 clusters per arm
If only 40 clusters (20 per arm): effective n = 20 x 200 / 10.95 = 365 sellers per arm
   -> detectable lift is only about 9.5pp, so a 3pp effect cannot be detected with simple randomization.
Continuous (GMV/seller, sd 150k, MDE 2.5k) at individual level: 56,512 per arm, before the DE inflation.
```

## Guardrails

- Seller churn and listing quality (sellers gaming volume to offset fees).
- Buyer conversion and search quality: supply changes affect the other side.
- Spillover: sellers active in both treated and control clusters.

## Decision rules (pre-committed)

| Outcome | Action |
|---|---|
| Activation up, net revenue per seller ≥ 0, supply healthy | Extend to more clusters |
| GMV up, net revenue down | Elasticity too low to pay for the discount: reject or narrow to high-elasticity segments |

| Outcome | Action |
|---|---|
| Parallel trends fail pre-period | Don’t trust DiD: use synthetic control or matched design |

## Likely follow-ups and strong answers

| Question | Answer |
|---|---|
| Why not randomize sellers? | Sellers compete for the same buyers and talk to each other: interference, fairness issues, and contaminated control. |
| Only 40 clusters. What do you do? | Matched-pair randomization, pre-period covariates (CUPED-like), DiD/synthetic control, accept a larger MDE, and run longer. State the detectable effect honestly. |
| Why is a hard A/B on price rarely used? | Leakage across units, legal/fairness constraints, and one-way trust damage. Prefer cluster tests, staggered rollouts, and natural experiments. |
| How do you estimate elasticity? | Combine DiD on historical changes, RD at fee-tier thresholds, and IV if a clean instrument exists. Segment sellers by size. |

# Project 6 · Netflix: home page to increase subscriptions (and a price increase)

## Context and assumptions

- **Part A (home page):** a redesigned landing/home experience for logged-out and free-trial visitors to raise visitor → paid subscription conversion.
- Baseline conversion 4%. 500,000 eligible visitors per week. New-subscriber month-2 retention 85%. Hours streamed in first 7 days: mean 8, σ 10.
- **Part B (price increase):** a subscription price rise in some markets. Individual-level price randomization is usually avoided (fairness, leakage, regulation), so use market-level tests.

## Hypothesis

A: If we show a clearer content-led home page to new visitors, paid conversion rises because they see value before the paywall, without lowering the quality of subscribers (retention and engagement).
B: A price increase raises revenue per member (ARPU) by more than it loses in subscribers; revenue-optimal if price elasticity $|\varepsilon|$ is low enough.

## Metrics

| Role | Metric | Type | Definition |
|---|---|---|---|
| Primary (A) | Visitor → paid conversion | Binary | Paid subscribers within 7 days of first visit / eligible visitors exposed |
| Diagnostic | Scroll depth, title clicks, sign-up start rate | Binary/Ratio | Funnel steps, computed per visitor |
| Continuous (A) | Hours streamed in first 7 days | Continuous | Per new subscriber: quality-of-subscriber proxy |
| Guardrail (A) | Month-2 retention of new subs; support contacts; page latency | Binary/Cont. | Don’t buy sign-ups that churn |
| Primary (B) | Net revenue per member | Continuous | ARPU × retained members, 90 days, market level |
| Guardrail (B) | Churn; downgrades to ad tier; sentiment/complaints | Binary | Price sensitivity and brand health |

## Design

| Item | Choice |
|---|---|
| Unit (A) | Visitor ID/device before login, stitched to account at sign-up; household-level to avoid people seeing both variants. Expect some cross-device leakage; check SRM |
| Entry (A) | First landing/home page view |
| Split (A) | 50 : 50 |
| Analysis (A) | Two-proportion z-test on conversion; Welch t on hours; month-2 retention at day 30 post-signup |
| Design (B) | Market-level: matched markets, DiD or synthetic control; staggered rollout; elasticity $\varepsilon$ = %Δ members / %Δ price |
| Duration (A) | ≥ 2 weeks of enrollment (week-to-week patterns) + 30-day retention window |

## Sample-size math

```
A. Conversion 4% -> 4.2% (+0.2pp): n = 154,302 per arm
   Traffic 500,000/week at 50/50 = 250,000 per arm per week -> 0.62 weeks for the primary; run >= 2 weeks anyway.
   Retention guardrail: month-2 85% vs 84% (-1pp harm): n = 20,557 new subscribers per arm.
   New subs per arm per week = 250,000 x 4% = 10,000 -> 2.06 weeks of enrollment -> 3 weeks to be safe (guardrail bin
   Hours streamed (sd 10, MDE 0.3h): n = 17,442 subscribers per arm.
B. Market-level: few units, so use DiD with pre-period fit; power is driven by the number of markets, not users.
```

## Guardrails

- Subscriber quality: month-2 retention and hours streamed; sign-ups that churn are not a win.
- Latency and page performance of the new home page.

- For pricing: churn, plan downgrades (ad tier), and revenue per member, not just subscriber count.

## Decision rules (pre-committed)

| Outcome | Action |
|---|---|
| A: conversion up, retention/hours flat or up | Launch; keep a 5% holdout |
| A: conversion up, month-2 retention down | Low-quality sign-ups: iterate (e.g. clearer pricing/value messaging) and rerun |
| B: net revenue per member up, churn within limit | Roll out in stages with continuous monitoring and DiD tracking |
| B: churn spikes or downgrades dominate | Roll back or soften (price-lock, promos for at-risk segments) |

## Likely follow-ups and strong answers

| Question | Answer |
|---|---|
| Why not run the price test at user level? | Customers compare prices (screenshots, social media), legal/fairness issues, and a one-way trust cost. Use market-level tests with DiD/synthetic control. |
| Short-term sign-ups up. Is that enough? | No: long-term value matters. Track month-2 retention, hours streamed, and LTV by variant before declaring success. |
| How do you handle logged-out users? | Randomize on a device/cookie ID, stitch to account at sign-up, and check SRM; accept some leakage and say so. |
| How do you decide the price change worked? | Net revenue per member (ARPU × retained members) vs. a matched control over 90+ days, with churn and downgrade guardrails; compute elasticity. |

# Part 5 · Question bank

Short, spoken-length answers. Add one example from your own work to each.

| Question | Answer |
|---|---|
| **Design** | |
| What is an A/B test and why run one? | Randomized control vs. treatment on a pre-defined metric. Randomization makes the groups comparable, so we get causation, an effect size, and side effects, not just correlation. |
| When would you NOT run one? | Obvious fixes (a broken checkout), too little traffic, ethical/legal limits, interference between units, or irreversible changes. Use DiD, synthetic control, RD, or staged rollouts instead. |
| How do you pick the unit of randomization? | The level at which the treatment is delivered and outcomes are measured; user/account by default, cluster when units interfere (sellers, friends, geos). Randomize at the unit you analyze, or adjust SE. |
| What is dilution and how do you avoid it? | Including users who never saw the change shrinks the measured effect. Enter users at first exposure, or analyze only exposed/eligible users. |
| How long should a test run? | n ÷ daily eligible entrants, ≥ 2 full weeks, plus the outcome window. Long enough for novelty to fade and cycles (weekday, salary, purchase) to be covered. No peeking. |
| What is an A/A test for? | Validates randomization, logging, and the false-positive rate (~5% of A/A tests should be significant); gives real variance for power. |
| What if assigned ≠ treated? | Analyze ITT (by assignment) for an unbiased effect; optionally estimate the effect on compliers with IV (CACE). |
| How do you ensure the sample represents the population? | Randomly assign within the eligible population that matches the rollout, cover full business cycles, check covariate balance, and check segment effects. |
| **Statistics** | |
| Explain the p-value to a PM. | “If the change did nothing, how often would we see a difference this big by chance?” It is not the probability the feature works or the size of the effect. |
| What does a 95% CI mean? | The procedure captures the true effect 95% of the time; a plausible range for the lift. If it excludes 0, significant at 5%. Report it with the effect size. |
| Type I vs. Type II error? | Type I ($\alpha$) = false positive: ship a dud. Type II ($\beta$) = false negative: miss a real win. Power = $1 - \beta$. |
| What is MDE and who sets it? | The smallest lift worth detecting; a business call (what lift pays back the cost), not a statistical one. |
| t-test vs. z-test? | z when $\sigma$ is known or n is large; t when $\sigma$ is estimated. They converge with n. Use Welch’s t for means and z/chi-square for proportions. |
| One-tailed vs. two-tailed? | Two-tailed detects harm too and is the default; one-tailed only if pre-registered and a negative result triggers the same action as a null. |
| What is CUPED? | Adjust the outcome with its pre-experiment value to cut variance by $(1 - \rho^2)$, so you reach power with fewer users. |
| How do you handle multiple metrics/variants? | One primary metric. For several comparisons control FDR (Benjamini–Hochberg); Bonferroni is conservative. |
| **Metrics** | |
| Primary vs. guardrail vs. diagnostic? | Primary decides; guardrails must not worsen; diagnostics explain the movement. One primary, 2–4 guardrails, a few diagnostics. |
| What makes a good metric? | Aligned to the goal, sensitive, reliable, hard to game, measurable in the test window. |
| Leading vs. lagging? | Leading (acceptance, early engagement) predicts; lagging (retention, LTV) confirms. Use leading metrics to read faster, lagging ones to validate. |
| Vanity metric example? | Total sign-ups or raw posts: they can rise while value falls. Prefer incremental, cohort-based metrics. |
| Ratio metrics: pitfall? | Randomize by user but divide by sessions → SE too small. Use the delta method or per-user aggregation. |
| **Judgment and scenarios** | |

| Question | Answer |
| --- | --- |
| p = 0.06. What now? | Not significant at 0.05 but not “no effect.” Check CI and power. Don’t extend to chase significance; rerun properly powered, or discuss the risk with stakeholders. |
| Primary up, guardrail down? | Don’t ship as-is. Investigate the cause, check segments, iterate and rerun. |
| Metric degrades mid-test? | Novelty effect first; also SRM, bug, seasonality. Look at the within-test trend. |
| SRM fails. What do you do? | Don’t read results. Find the cause (assignment bug, filters, crashes, logging), fix, rerun. |
| Network effects? | Cluster/graph randomization, geo tests, or switchback. Trade-off: less power. |
| Can’t randomize? | DiD (parallel trends), synthetic control, RD, PSM (weakest), CausalImpact; state the assumptions. |
| Stat sig but tiny effect? | Practical significance: check effect size, CI, and ROI before shipping. |
| PM wants to ship without a test? | Offer a lighter alternative: staged rollout with a holdout and pre-agreed guardrails, or a DiD, not a flat “no.” |

# Part 6 · The 60-second answer and checklist

## 60-second script for “Walk me through how you’d A/B test X”

“First I’d clarify the **decision** this informs and write a **hypothesis**: if we show X to Y users, Z changes because... My **primary metric** is [defined], with [2–3] **guardrails** and a few diagnostics. I’d randomize at the **user** level and enter users at first exposure to avoid dilution. With α 0.05, power 0.8, a baseline of [p] and an MDE of [δ, justified by business value], I’d size [n per arm], which at [traffic] means [N] days, rounded up to full weeks. I’d check **SRM** before reading results, watch for novelty and seasonality, analyze with the right test for the metric type, and report **effect size with a CI**. I’d pre-commit decision rules: launch if the primary clears the MDE and guardrails are flat, iterate if mixed, stop if flat. A holdout stays on for long-term effects.”

## Final checklist

- Decision and hypothesis stated with a mechanism.
- One primary metric defined with numerator, denominator, and window; guardrails with thresholds.
- Unit, eligibility, and entry point justified; interference considered.
- α, power, baseline, MDE, split, n, and duration given with numbers.
- Metric type → right test; skew/ratio/cluster issues addressed.
- SRM, novelty, seasonality, and concurrent tests checked.
- Effect size and CI reported; ITT; segments examined.
- Decision rules pre-committed; long-term holdout planned.
- One real example from your work for each step (Meta incrementality holdout, festive churn holdout).

## Numbers worth remembering

| Quantity | Value |
|---|---|
| Z (α=0.05 two-sided) / Z (power 0.8) | 1.96 / 0.84, so $(Z+Z)^2 = 7.85$ |
| n per arm, binary | $\approx 15.7 \cdot p(1-p) / \delta^2$ |
| n per arm, continuous | $\approx 15.7 \cdot \sigma^2 / \delta^2$ |
| Halve the MDE | ≈ 4× the sample |
| CUPED | variance × $(1 - \rho^2)$ |
| Design effect | $1 + (m - 1) \cdot ICC$ |
| 20 : 80 split vs 50 : 50 | ≈ 1.6–1.7× traffic |